PRABHAT RANJAN JHA

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EDUCATION

Scaler School of Technology — B.Tech, Computer Science *2025 – 2029*

Indian Institute of Technology, Madras — B.S. Data Science & Applications (Online) *2025 – 2029*

Shiv Vani Model Senior Secondary School — HSC (87.4%) *2024*

EXPERIENCE

Web Development Teaching Assistant — Scaler School of Technology *Aug 2026 – Present*

- Mentor 36 first-year students 1:1 on HTML, CSS, JavaScript, React, and databases, resolving conceptual and debugging doubts.
- Focus on the 'why' behind each concept so mentees can reason through problems instead of memorizing syntax.

PROJECTS

Relayo — AI Automation System for a Messaging Startup *n8n, Groq LLMs, Google Sheets, Gmail, Discord API, Slack*

- Built 4 independent n8n workflow modules (email, billing, Discord, self-monitoring) that classify each item, draft a reply, then auto-send or flag it for human review.
- Decoupled review from sending: flagged items are logged to Google Sheets and a separate scheduled workflow sends them once approved, so no execution sits idle waiting on a human.
- Hardened the email handler with strict-JSON LLM output, Gmail message-ID dedupe, silent spam discard, and phishing and prompt-injection defenses; billing disputes always go to a human and invoices are templated, never LLM-generated.
- Worked around n8n's missing Discord trigger by polling the REST API with workflow static data; added watchdogs for reviews stale 3+ days and a health check on Groq, Gmail, Sheets, and Discord.

Ripple — Project Planner with Critical-Path Analysis *MERN, Mongoose, JWT, Tailwind CSS*

- Built a full-stack planner that turns tasks, durations, and dependencies into a schedule and critical path, with a what-if simulator showing how a slipped task moves the finish date without touching the real plan.
- Implemented JWT auth in HttpOnly cookies with bcrypt hashing, and configured cookie and CORS settings separately for development and production.

Nexora — Full-Stack E-Commerce Platform *React 19, Vite, Express, MongoDB, Tailwind CSS*

- Built customer auth (JWT in HTTP-only cookies, protected and public routes via React context) and profile management with shipping address and image upload (Multer, 2 MB limit).
- Built product catalogue APIs with case-insensitive search, category filtering, and price sorting, plus a persistent per-customer wishlist behind protected REST endpoints.

TA Doubt Assistant — RAG-Powered Support Bot *n8n, LangChain, HuggingFace, FAISS*

- Built a RAG workflow with semantic search that auto-answers student doubts, remembers follow-ups, escalates unresolved queries by email, and grows its own knowledge base.
- Re-implemented each n8n concept (tool calling, conversation memory, RAG) in LangChain (Python) to understand the underlying mechanics.

OTHER PROJECTS

- **Blood Donor Finder** (React, Hooks, REST API) — search and filter donors with live availability sorting and a request-help flow.
- **Gesture Control** (Python, OpenCV) — hand-gesture game controller, including a custom build for ev.io.
- **Orisca** (HTML, CSS) — space-themed portfolio with 3D rotating navigation and custom CSS animations.
- **CodeQuest** and KeyMonk (HTML/CSS/JS) — a visual DSA explainer site and a Monkeytype-style typing test with live WPM/accuracy.

TECHNICAL SKILLS

Languages: JavaScript, Python, Java, HTML5, CSS3

Web (MERN): React, Node.js, Express, MongoDB/Mongoose, REST APIs, JWT auth, Tailwind CSS, Vite

Automation & AI: n8n, LangChain, RAG (FAISS, HuggingFace embeddings), LLM tool-calling (Groq), Slack/Gmail/Google Sheets/Discord integrations

ML (coursework): Regression, kNN, Decision Trees, Ensembles, Clustering, PCA

Tools: Git, GitHub, OpenCV

ACHIEVEMENTS & ACTIVITIES

- 3rd Place (Team) — RoboDog Locomotion Challenge Hackathon
- 3rd Place — Escape Room Competition, Paradox Fest, IIT Madras
- Participant — Auraverse & Auraverse 2.0 Hackathons, Scaler School of Technology
- Volunteer — Meta x Scaler Hackathon